<Exam Teams use – File Name>

HERIOT WATT UNIVERSITY

SCHOOL OF SOCIAL SCIENCES

Edinburgh Business School

C21AO and C27IE

Economics / Introductory Economics

IN-PERSON EXAMINATION

December Examination and Assessment Diet 2023

Duration: 2 hours

There are three sections to the examination paper
In Section A, there are **20 multiple choice questions**. You should attempt **all 20 questions**. Each question is worth 1.5 marks.
Multiple choice answer grid should be detached from exam paper and attached to answer booklet.
In Section B, there are **four** questions. You should attempt **all four questions**. Each question is worth 10 marks.
In Section C, there are **five** questions. You should attempt **any two questions**. Each question is worth 15 marks.

Attached: Multiple choice answer grid
Supplied: One sixteen page examination booklet
Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

DO NOT REMOVE THIS PAPER FROM THE EXAM HALL

Section A

There are **20 multiple choice questions** in Section A. Please attempt **all these questions**, using the answer grid at the end of the examination, inserting the answer grid into your examination booklet when you are finished.

1) At the current market price of 120, there is excess demand for running shoes. We expect that:
   a) The market price will increase, the quantity supplied will increase, and the quantity demanded will decrease.
   b) The market price will decrease, the quantity supplied will increase, and the quantity demanded will decrease.
   c) The market price will decrease, the quantity supplied will decrease, and the quantity demanded will increase.
   d) The market price will increase, the quantity supplied will decrease, and the quantity demanded will increase.

2) Which of the following is not a problem to which we can easily apply microeconomic analysis?
   a) Understanding the response of firms to an increase in the wages which they must pay their workers.
   b) Understanding how the market price and firm output will change when new businesses enter a competitive market.
   c) Identifying a suitable government response to an energy price shock, which causes both unemployment and inflation.
   d) Estimating the margin over the competitive market price which a firm with market power can charge.

3) Which of the following are characteristics of the market for an abundant good?
   a) There will be excess supply of the good.
   b) The good will be freely available to all potential users of it.
   c) People will not face any limits on their use of the good.
   d) All three statements in parts a) – c).

4) Coca Cola produces many sugary drinks, including Sprite and Fanta. Assuming that there is limited substitutability between Sprite and Fanta, we expect that the cross price elasticity of demand will be:
   a) Less than -1.
   b) Between -1 and 0.
   c) Between 0 and 1.
   d) More than 1.

5) A fashionable cookery writer recommends using mangos in dessert in a viral Tiktok at the same time as trade unions negotiate higher wages for workers. We expect that:
   a) The demand curve and the supply curve will both shift to the left.
   b) The demand curve will shift to the right and the supply curve will shift to the left.
   c) The demand curve will shift to the left and the supply curve will shift to the right.
   d) The demand curve and the supply curve will both shift to the right.

6) In many developing countries, cars are superior goods. In these countries:
   a) The income elasticity of demand for cars is greater than one.
   b) The income elasticity of demand is less than zero.
   c) The price elasticity of cars is less than minus one.
   d) The price elasticity of cars is greater than zero.

7) Roots and Fruits operates in a monopolistically competitive market. Analysing its costs of production, we expect to find that:
   a) Its average variable costs are greater than its average costs.
   b) Its average fixed costs increase when they are less than its marginal costs.
   c) Its average variable costs are at their minimum when they are equal to its marginal costs.
   d) Its marginal costs continue to fall as it expands its output.

8) KTM is the main train operator in Malaysia, and so has substantial monopoly power. As a result, we expect that:
   a) It will restrict services to increase its profits.
   b) It will not be able to generate long-run economic profits.
   c) It will adopt marginal cost pricing to maximise its profits.
   d) It will set lower train fares than train operators in competition with each other.

9) Driving into work, when traffic slows down on the motorway because there is traffic congestion, this is evidence of:
   a) The road network being a public good.
   b) The government regulating an externality efficiently.
   c) Driving on the motorway being a non-rivalrous activity.
   d) Access to the road system being non-excludable.

10) The market for mobile phones is dominated by a small number of large businesses, so has the form of an oligopoly. We expect that:
    a) These firms need to accept the market price.
    b) These firms may be able to collude to raise the market price.
    c) The market price will be lower than if the market were to become more competitive.
    d) The market price will ensure that firms only make normal profits, and break even.

11) After the government imposes a tariff on imports of steel, which group can we be certain will be worse off?
    a) Firms which use steel in production.
    b) The Government.
    c) Domestic firms which produce steel.
    d) Workers employed in the steel industry.

12) If the Malaysian ringgit appreciates against the pound sterling, then:
    a) Malaysian students in Edinburgh will find that the cost of their tuition fees (in ringgits) has increased.
    b) When the University makes payments from Malaysia to Edinburgh, it will have to pay more (in ringgits).
    c) There will be inflationary pressures in Malaysia.
    d) Malaysia will find that demand for its goods in the UK has fallen.

13) Which of the following is not a basis for calculating the GDP measure of national income?
    a) The expenditure on all final goods and services produced within the economy.
    b) The incomes received by households from the provision of factors of production.
    c) The transfers which the government makes to support disadvantaged families.
    d) The value added in the production of goods and services.

14) Why might the multiplier effect on income of a reduction in taxation be less than the multiplier effect of an increase in government expenditure?
    a) Government can only spend money on domestic goods and services.
    b) Households are likely to save some of the reduction in taxation.
    c) Businesses charge the government higher prices than households.
    d) There will be more investment after an increase in government spending.

15) Which statement best captures the extreme Keynesian beliefs about the shape of the aggregate supply curve?
   a) The long run aggregate supply curve is upward sloping, but steeper than the short run aggregate supply curve.
   b) The long run aggregate supply curve is horizontal.
   c) The short run aggregate supply curve is horizontal up to the capacity of the economy, and then becomes vertical.
   d) The short run aggregate supply curve is vertical.

16) When the central bank sells short term government bills to the commercial banks:
   a) The government is reducing its use of deficit financing.
   b) The central bank is reducing the money supply.
   c) The central bank is using monetary policy to reduce interest rates.
   d) The government is running an expansionary monetary policy.

17) If taxation is a quarter of income, there is no saving, and one third of disposable income (income after tax has been paid) is spent on imports, then the marginal propensity to spend on domestic goods is:
   a) 0.75
   b) 0.67
   c) 0.5
   d) 0.33

18) Which of the following is a supply side policy?
   a) Reducing the demand for credit by increasing the discount rate of interest on lending by the central bank to commercial banks.
   b) Reducing regulations associated with hiring workers and terminating their contracts.
   c) Increasing government spending, with the deficit funded by the issue of long-term bonds.
   d) Increasing the money supply with the central bank purchasing government bills held by commercial banks.

19) Which of following is a statement of the natural rate hypothesis?
   a) There is level of unemployment at which the rate of inflation will remain static.
   b) There is a level of national income at which there will be full employment.
   c) There is a level of growth of national income which is consistent with macroeconomic stability.
   d) There is a level of the growth of the money supply which ensures that there will be persistent economic growth.

20) When there is an increase in investment, increasing the productive capacity of the economy, we might capture the effects in an aggregate demand-aggregate supply diagram with:
   a) A shift of the aggregate demand curve to the left
   b) A shift downwards of the short-run aggregate supply, while the long-run aggregate supply curve does not move.
   c) A downward shift of the aggregate demand curve, while the long-run aggregate supply curve becomes flatter.
   d) A shift to the right of both the aggregate demand curve and the long-run aggregate supply curve.

(Total: 30 marks)

Section B

There are **four short answer questions** in this section, all of which require you to carry out some calculations. You should answer **all four questions**. Each question is worth 10 marks.

1) Crystal Clear Audio makes headsets which can be used with laptop computers. It has estimated demand for its output at prices from 10 up to 60.

Price	0	10	20	30	40	50	60
Output (Thousands)	-	3,600	1,800	1,200	900	720	600

a) Define the concept of the price elasticity of demand, and explain how to measure it.

(2 marks)

b) Using the mid-point formula for the price elasticity of demand, calculate the price elasticity of demand for headsets when the price changes from 10 to 20, and also when it changes from 10 to 60.

(4 marks)

c) Comment on the nature of the price elasticity of demand, and how demand changes as price changes. Without doing any further calculations, use the value of the price elasticity of demand to state what will happen to Crystal Clear Audio's sales revenue as it increases its prices, and the value of its marginal revenue.

(4 marks)

(Question 1 Total: 10 marks)

2) Crystal Clear Audio now diversifies into the production of video cameras. It has estimated how the number of cameras which it could produce in a year will change with the number of workers which it hires.

a) Define the average physical product and the marginal physical product. Complete the table below.

(4 marks)

Workers	0	20	40	60	80	100	120
Output (Thousands)	0	50	130	200	250	270	250
APP							
MPP							

b) Sketch a diagram showing the average physical product and the marginal physical product. In your sketch, show clearly where the average physical product and the marginal physical product are equal.

(4 marks)

c) Assume that Crystal Clear Audio pays its workers 30,000. Given that wages are the only variable costs, complete the table below, showing the average variable cost of output.

(2 marks)

Output (thousands)	0	50	130	200	250	270	250
TVC (thousands)							
AVC							

(Question 2 Total: 10 marks)

3) You are to analyse this model of aggregate demand.
For national income (GDP) *Y,* aggregate consumption, *C*, savings, *S*, investment, *I*, government expenditure, *G*, taxation *T*, income after taxation, Y^D, exports, *X* and imports, *M*, where

$C = 0.8Y^D$
$T = 0.25Y$
$M = 0.2Y$
$G = 1{,}000$, $I = 500$, $X = 900$

a) State an equilibrium condition, give an economic interpretation of it, and then use it to find the equilibrium level of national income, Y^*

(5 marks)

b) Define the government expenditure multiplier, explain why it typically has a value which is greater than one, but finite, and calculate its value in this case.

(5 marks)

4) Atlantis and Oceania produce two fruits, melons and figs. Each has a workforce of 25,000 people. In the table below, you can see the quantities which each country would produce were the whole of its workforce engaged in the production of each fruit. You may assume that as workers shift between production of figs and melons, the marginal product of both industries remains constant.

Capacity (000 tonnes)	Country	
	Atlantis	Oceania
Melons	200	300
Figs	50	60

Without trade, both countries choose to produce a combination of both fruits. Atlantis produces 120,000 tonnes of melons and 20,000 tonnes of figs; and Oceania produces 150,000 tonnes of melons, and 30,000 tonnes of figs.

a) Define the concept of *absolute advantage*. State which country has the absolute advantage in the production of each fruit.

(2 marks)

b) Define the concept of comparative advantage. Explain why it is impossible for either country to have a comparative advantage in the production of both goods, and state for the production of which good each country has a comparative advantage.

(3 marks)

c) The fig farmers of Atlantis have claimed that it is possible that if the country gives up melon production, and trades with Oceania, it will be possible to consume more figs and more melons. Examine this claim.

(5 marks)

(Question 4 Total: 10 marks)

Section C

There are **five discussion questions** in this section, each of which is worth **thirty marks**. You should attempt any **two questions**.

1) Complete a PEST analysis of the market environment facing an international supplier of higher education, which is considering setting up business in a new country.

(15 marks)

2) In many countries, we find that there is dominant supplier of energy for domestic use, which has substantial market power.

a) Applying the Structure-Conduct-Performance approach, explain the likely differences between a monopoly and a competitive industry.

(6 marks)

b) Using a diagram, explain why we expect that a profit-maximising monopoly will produce less output, and charge a higher price than if it were to produce the socially optimal output.

(6 marks)

c) Briefly explain why we may prefer to have a monopoly supplier than insist upon competition in markets.

(3 marks)

(Question 2 Total: 15 marks)

3) In economic analysis, we consider climate change to be one of the most important examples of a negative externality. Explain how we analyse the effects of an externality, and set out how this approach might be applied to the case of climate change.

(15 marks)

4) For many years, economists have used the term 'business cycle' to refer to the succession of periods of economic expansion, followed by recession, over approximately 7 – 9 years. An important objective of government economic policy is to manage these fluctuations.

a) Briefly outline the links between the business cycle and the credit cycle in banking.

(6 marks)

b) How might governments design their policies so that measures to reduce the fluctuations in the business cycle either occur automatically or else follow rules?

(9 marks)

(Question 4 Total: 15 marks)

5) The exchange rate between the US dollar (USD) and the Euro (EUR) changes every day. We define it as the number of Euros required to buy one US dollar.

a) List the four sources of variability in the exchange rate, briefly stating their effect.

(8 marks)

b) Discuss how exchange rate variability might affect the government's ability to support economic growth.

(7 marks)

(Question 5 Total: 15 marks)

END OF PAPER

On page 14, you will find the multiple choice answer grid.

Section

Answer Grid for the multiple choice questions

Please place a cross in a box to indicate clearly your chosen response to each question. If you wish to change your response, score through the box which you had chosen first, and then place a cross in the box corresponding to your preferred response.

If you leave crosses in more than one box, you will obtain no credit.

Multiple choice answer grid should be detached from exam paper and attached to answer booklet.

	Answer			
Question	A	B	C	D
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